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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – MARCH 2024

Program	:	**B.E. - CSE (AI & ML) / CSE (CY) / AI & ML / AI & DS**	**Semester**	:	**III**
Course Name	:	**Discrete Mathematical Structures**	**Max. Marks**	:	**100**
Course Code	:	**CI35 / CY35 / AI34 / AD34**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

1. a) Construct the truth tables for the following compound statement $\neg(p \vee (q \wedge r)) \Leftrightarrow (p \vee q) \wedge (p \rightarrow r)$. CO1 (06)

   b) Prove that the number $\sqrt{2}$ is irrational by method of contradiction. CO1 (07)

   c) Using Mathematical induction, prove that for any positive integer n, 7 divides $3^{2n+1} + 2^{n+2}$. CO1 (07)

2. a) Prove that if $n = ab$ where a, b are positive integers then $a \le \sqrt{n}$ or $b \le \sqrt{n}$ by contrapositive method. CO1 (06)

   b) Without constructing truth tables, prove that $[\neg p \wedge (\neg q \wedge r)] \vee [(q \wedge r) \vee (p \wedge r)]$ is logically equivalent to r. CO1 (07)

   c) Write the converse, inverse and contrapositive for "If a quadrilateral is a parallelogram, then its diagonals bisect each other". CO1 (07)

UNIT – II

3. a) Define Boolean Algebra. Verify whether $(S_{42}, /)$ is a Boolean Algebra or not? CO2 (06)

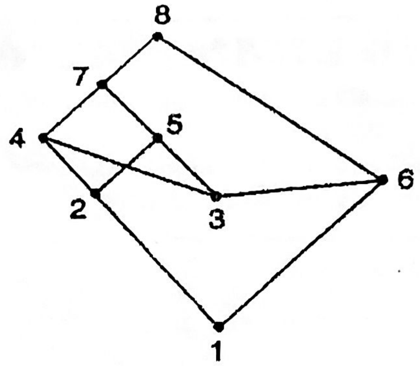
   b) Let $A = \{a, b, c, d\}$ and consider the relation $R = \{(a,a), (a,b), (a,c), (a,d), (b,b), (b,d), (c,c), (c,d), (d,d)\}$. Show that R is a partial order relation. Hence draw Hasse diagram. CO2 (07)

   c) Solve the recurrence relation $a_n = a_{n-1} + 2, n \ge 2$ subject to the initial condition $a_1 = 3$. CO2 (07)

4. a) Let $A = \{1,2,3\}$ and R, S be the relations on A whose matrices are as given below. Find the relations $\overline{R}$, R^{-1} and their matrix relations. Also draw their digraphs. CO2 (06)

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad M_S = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

[redacted] b) F[redacted]m[redacted]the following Hasse diagram, find maximal elements, minimal elements, upper bounds, lower bounds, lub and glb of $A = \{2,3\}$ and $B = \{4,6\}$ if exists. CO2 (07)

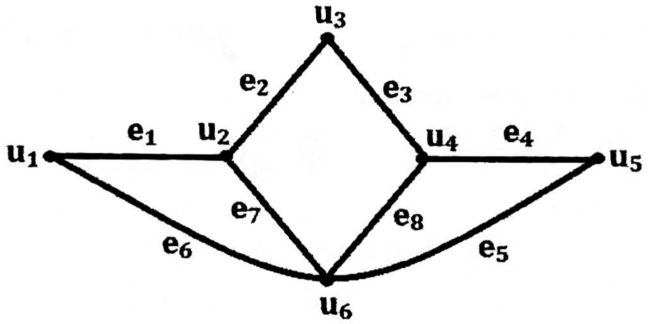


c) Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}, n \geq 2$ subject to the initial conditions $a_0 = 0, a_1 = 3$. CO2 (07)

UNIT - III

5. a) Define (i) walk (ii) Path (iii) Circuit. CO3 (06)

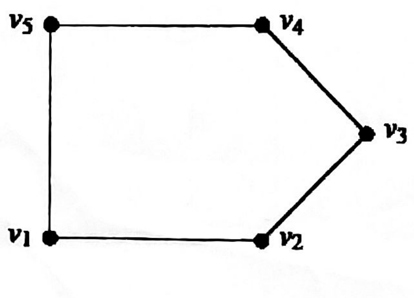
b) Define fusion of two vertices of a graph. Find (i) Fusion of u_1 and u_2 (ii) $G - e_1$ (iii) $G - u_1$. for the graph. CO3 (07)

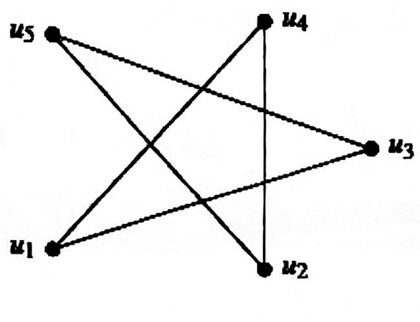


c) Draw the graph whose incidence matrix is CO3 (07)

$$M = \begin{array}{c} \\ u \\ v \\ w \\ x \\ y \end{array} \begin{bmatrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

6. a) Define isomorphism of two graphs.Check whether the following graphs are isomorphic or not. CO3 (06)

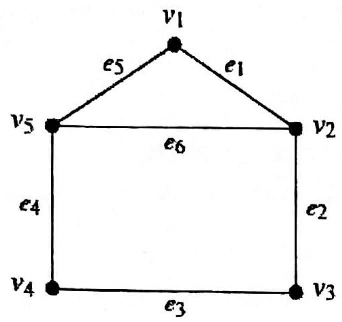




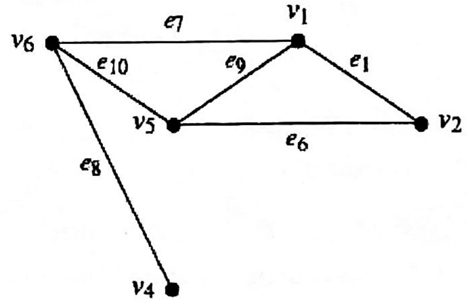
b) Draw the graph whose adjacency matrix is and hence write its incidence matrix. CO3 (07)

$$X(G) = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

c) Find $G_1 \cap G_2$, $G_1 \oplus G_2$ of CO3 (07)



G_1



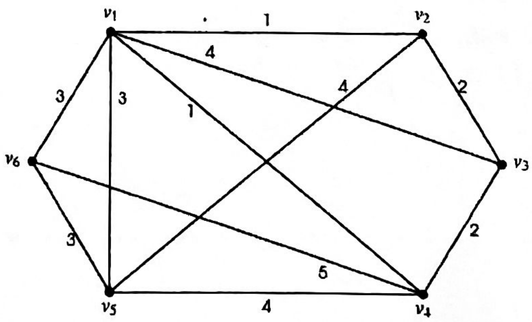
G_2

UNIT- IV

7. a) Define (i) spanning tree (ii) minimally connected graph with an example (iii) cut-sets with an example. CO4 (06)

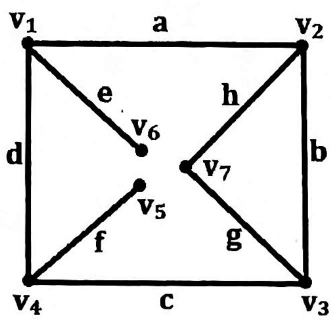
b) (i) Prove that in a tree, there is one and only path between every pair of vertices. CO4 (07)
(ii) Prove that a tree with n vertices has $n-1$ edges.

c) Explain Kruskal's algorithm to find the shortest spanning tree and hence find the minimal spanning tree in each of the following graphs. CO4 (07)

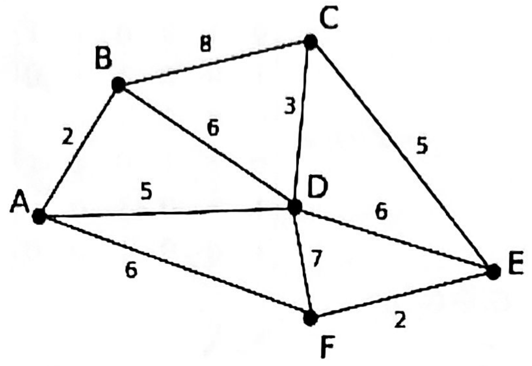


8. a) Prove that a graph G is a tree if and only if it is minimally connected. CO4 (06)

b) (i) Define cut-set matrix and weighted graph. CO4 (07)
(ii) Find cut-sets of the following graph.



c) Explain Prim's algorithm to find the shortest spanning tree and hence find the minimal spanning tree in each of the following graphs. CO4 (07)



UNIT - V

9. a) Define (i)Semi group (ii) Monoid with example for each. CO5 (06)

b) A non-empty subset H of a group G is a subgroup G if $a,b \in H \Rightarrow ab^{-1} \in H$ where b^{-1} is inverse of b in G. CO5 (07)

c) Let G be a group of positive real numbers under multiplication and G' be a group of all real numbers under addition .Let $f:G \to G'$ such that $f(x) = \log_{10}^{x}$ for $x \in G$.Show that $f:G \to G'$ is isomorphism CO5 (07)

10. a) Define Homomorphism and Isomorphism. CO5 (06)
If $G = \{1,-1,i,-i\}$ is a group under multiplication then write all subgroups of G.

b) Let Z^{+} be the set of all positive integers and * is an operation defined by $a * b = Max\{x, y\}$ for $a,b \in z^{+}$.Verify $(z^{+},*)$ is Monoid or not. CO5 (07)

c) A non-empty subset H of a group G is a subgroup G if CO5 (07)
(i) $a \in H, b \in H \Rightarrow ab \in H$
(ii) $a \in H \Rightarrow a^{-1} \in H$.

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